

Aryan Isaac Bhobe

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Education

University of Bath

Oct 2023 – Jul 2027

B.Sc (Hons) Computer Science & Artificial Intelligence

Bath, United Kingdom

- Classification – **First Class Honours (1:1)**
- Relevant Courses – Finance & Risk Management, Bayesian Machine Learning, Reinforcement Learning, Natural Language Processing, Algorithms & Data Structures, Probability, Statistics, Linear Algebra, Software Engineering, Databases, Business Strategy

Experience

Accenture

Jun 2026 – Sep 2026

AI & Data Summer Analyst

London, United Kingdom

- Building production agentic AI pipelines on **GCP** using **Google ADK** with agentic RAG over a **Supabase** vector store; developed and configured **MCP servers** to expose enterprise tools and data sources to agent runtimes, with services built in **FastAPI** deployed via **Cloud Run** with **Apigee** governance, supporting a major UK telecommunications client.
- Architected a network-isolated enterprise RAG system on Azure for a major energy client: **Azure AI Foundry** and **AI Search** over confidential SharePoint data, **Cosmos DB** for memory persistence, **React/FastAPI** on App Service; all inter-service traffic enforced via **VNet** integration and private endpoints; researched productionisation pathway from prototype to secure enterprise deployment.

Merck Sharp & Dohme (MSD)

Jun 2025 – Jun 2026

Machine Learning Scientist

London, United Kingdom

- Led independent ML research on a **550k-sample** dataset across 18 relational tables; applied unsupervised learning (**K-Means**, **GMM**, **HDBSCAN**) with **MCA** dimensionality reduction to uncover clinically meaningful patient subtypes on GPU-accelerated **HPC**.
- Designed a systematic model selection framework using **ARI** agreement between K-Means and GMM, k-sweep and hyperparameter optimisation via **BIC** and silhouette scoring; validated cluster stability via **bootstrap resampling** and cross-referenced against HDBSCAN density-based clustering.
- Engineered reproducible ML pipelines with versioned configs and modular code; produced clinical spider plots and enrichment heatmaps, presenting findings to senior leadership in the Neuroscience AI team.
- Deployed end-to-end proteomics QC pipelines adopted as the **departmental standard**: **PCA**-based outlier detection and **KNN/Random Forest** imputation on high-dimensional biological data, documented as a reproducible framework for biomedical agent training.
- Conducted **longitudinal modelling** across cohorts via regression, hierarchical clustering, and Jaccard similarity; applied hypothesis testing (t-test, chi-squared), contributing to a poster at the Boston neuroscience symposium.

Spookfish Innovations

Jul 2024 – Sep 2024

Deep Learning Engineer Intern

Bengaluru, India

- Led model selection across instance segmentation architectures (**YOLOv9**, **RTMDet**, **Mask R-CNN**) for a real-time ad-insertion system, configuring custom data loaders and augmentation strategies in PyTorch.
- Deployed real-time inference engine integrating **OpenCV** with PyTorch, implementing image warping and dynamic mask composition via bitwise operations for frame-accurate ad insertion.

Projects

Glencore Quant Research & Multi-Agent Platform | [Code](#) | [Demo](#)

Python | OpenAI Agents SDK | FastMCP

- Conducted quantitative analysis of Glencore (GLEN.L), demonstrating that the apparent annual price cycle was attributable to semi-annual **dividend adjustments** rather than market seasonality, via stationarity tests, spectral analysis, and calendar decomposition.
- Developed GJR-GARCH **volatility models** and commodity market analyses, confirming persistent volatility clustering and leverage effects while identifying copper prices as a statistically significant indicator of Glencore returns through **Granger causality** testing.
- Built an interactive quantitative research platform integrating walk-forward ML forecasting with an OpenAI multi-agent architecture (**Agents SDK + FastMCP**), enabling natural-language exploration of financial models through deployed analytical tools on **Streamlit**.

Equity Volatility Forecasting: GARCH vs ML | [Report](#) | [Code](#)

Python | statsmodels | XGBoost | SHAP

- Benchmarked econometric (GARCH family) and machine learning models for **1-day-ahead SPY volatility forecasting** over a 13-year period, motivated by applications in options pricing and Value-at-Risk estimation.
- Random Forest reduced forecasting RMSE by **14%**, while GARCH-t achieved the lowest QLIKE loss during crisis regimes, demonstrating the robustness of heavy-tailed volatility models for tail-risk estimation.
- Identified a regime inversion in model performance: machine learning models deteriorated by **33–58%** during COVID-19 stress periods while GARCH models became more reliable; SHAP analysis identified VIX as the dominant predictive feature.

CDS Spread Modelling with Support Vector Regression | [Report](#) | [Code](#)

Python | scikit-learn | pandas

- Developed an end-to-end Support Vector Regression framework for estimating 5-year CDS spread proxies for unrated firms using macroeconomic and equity market features, achieving $R^2=0.849$ and **RMSE=33.3 bps** on a temporally held-out test set.
- Optimised an RBF SVR model through 5-fold cross-validation across 625 hyperparameter combinations, implementing leakage-free preprocessing and target transformation to ensure robust out-of-sample evaluation.
- Demonstrated a **4.6× degradation** in forecasting accuracy during COVID-19 stress periods using expanding-window validation, highlighting regime dependence and model risk in production credit modelling.

Skills

- **Languages** – Python, C++, SQL, JavaScript
- **AI/ML** – PyTorch, TensorFlow, scikit-learn, ADK, XGBoost, JAX, CNNs, Unsupervised Learning, LangGraph, MCP, Claude
- **Quantitative** – statsmodels, SciPy, NumPy, Pandas, SHAP, Matplotlib, Seaborn, Hypothesis Testing, Time Series
- **Tools** – GCP, AWS, FastAPI, Cloud Run, Apigee, Git, GitHub, MySQL, Redshift, MongoDB, Tableau, Excel, Supabase